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Research Paper

Influence of Sputtering Pressure on the Conductivity and Transparency of Aluminum-Doped Zinc Oxide Films

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ABSTRACT

Aluminum-doped zinc oxide (AZO) films are promising candidates for transparent electronics due to their low resistivity, high transmittance, and long-term stability. In this study, we investigated the impact of sputtering pressure on resistivity saturation in AZO films that are deposited by using radio frequency magnetron sputtering on transparent glass at room-temperature (RT). An X-ray diffraction analysis reveals that RT-deposited AZO (RT-AZO) films prepared at a pressure of 20 Pa exhibit a predominant orientation along the *a*-axis, favoring the (100) direction, as well as the lowest resistivity value of ~ $4 \times 10^{-3}\,\Omega$·cm and the highest transmittance (95 %) in the visible range. These findings highlight the potential of our deposition approach for producing highly oriented (100) RT-AZO films, paving the way for their application to low-cost transparent electronics.

Keywords: Aluminum-doped zinc oxide, Radio frequency magnetron sputtering, Room-temperature, Transparent electronics

1. Introduction

The optics and photonics sectors have traditionally utilized transparent conducting oxides (TCOs) such as indium tin oxide (ITO) for various applications involving solar cells and transparent thin-film transistors [1]. However, the use of ITO currently faces challenges such as high costs, supply unpredictability due to limited indium deposits, and reduced durability from plasma damage [2]. Doped zinc oxide (ZnO) films could be a potential solution to these issues because they are less expensive, non-toxic, and more stable than ITO films, and they also have better electrical and optical properties [3]. For example, aluminum (Al) or aluminum oxide (Al_2O_3) doped ZnO (AZO) films exhibit high conductivity and a large optical band gap of 3.3 eV [4]. Additionally, AZO films are unique in that they can be obtained in a crystalline form at ambient temperature, setting them apart from other materials [5–8]. Therefore, AZO has many potential uses, including in surface acoustic wave devices [9], gas sensors [10], light-emitting diodes (LEDs) [11], and low emissivity films [12]. In addition, AZO films have been used as front TCOs and back reflector coating for high-efficiency solar cells due to their high haze ratio for light scattering and light trapping [13].

Various deposition systems have been employed for the deposition of AZO films including a sol-gel process, radio frequency (RF)-magnetron sputtering, pulsed laser deposition, and electron beam evaporation [14–19]. Among them, RF-magnetron sputtering is a popular choice due to its low cost, simplicity, capability to deposit highly crystalline films that are uniform, dense, and strongly adherent, and low operating temperature [20,21]. However, achieving a high figure of merit in AZO films at ambient temperature, which could circumvent degradation of the sensitive photoresist or plastic substrates used in opto-electric devices (e.g., LEDs), remains a challenge. In addition, there is still need for a thorough investigation into how the properties of AZO films are affected by the sputtering pressure at ambient temperature.

In this study, we investigated the influence of growth pressure on the transparency and conductivity of AZO films sputtered at room-temperature (RT) AZO. We achieved highly oriented *a*-axis (100) RT-AZO films with high conductivity and transparency by deposition at a sputtering pressure of 20 Pa. In comparison to typical AZO films oriented along the *c*-axis of the (002) plane, RT-AZO films oriented along the *a*-axis are anticipated to exhibit a superior second harmonic conversion coefficient [22–25]. We achieved a transmittance of 95 % and a resistivity of ~ $4\times10^{-3}\,\Omega$·cm for RT-AZO without additional heat or laser treatment [26–32]. Our findings suggest that this deposition method has the potential for producing high-performance AZO films for optoelectronic applications.

2. Experimental details

RT-AZO films were deposited on bare glass (Corning Eagle 2000) using RF-magnetron sputtering with a 4-inch circular disk type Al_2O_3

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2 wt.% doped ZnO target. The sputtering pressure for RT-AZO deposition was adjusted within a range of 0.13 – 20.00 Pa while the substrate temperature was maintained at RT. Before the deposition process, the glass substrates were cleaned by an ultrasonication system in acetone and methanol for 10 min to eliminate surface impurities. The substrates were dried by blowing nitrogen gas of 99.99 % after being washed with deionized water. During the sputtering process, the glass substrates were positioned 5 cm from the AZO target.

The base pressure of the sputtering chamber was 1.0×10^{-4} Pa. Impurities or contaminants on the surface of the target were removed by performing a 10 min pre-sputtering process using a high purity (99.999 %) Ar gas. The RF power was kept at 150 W throughout the deposition process. The thickness of the RT-AZO films was measured by a spectroscopic ellipsometer (Nano-View, MF-1000), and the estimated average thickness of the as-deposited films was approximately 100 nm.

The structural characteristics of the RT-AZO films were examined by using X-ray diffraction (XRD) (Bruker D8 Discover), and the detailed setting values were as follows: Cu-Kα radiation source: 0.154 nm, angular scan speed: 0.1 °min^{-1}, and specific angle: 20 to 50°. The transmittance of the RT-AZO films was examined through an ultraviolet-visible spectrometer (SCINCO, S-3100). The electrical properties of the samples were evaluated with the aid of Hall measurement (ECOPIA, HMS-300). An energy dispersive X-ray spectroscopy (EDS, JEOL, JSM-6700F) analyzer was used to verify the chemical component of the as-prepared RT-AZO films.

3. Results and discussion

Figure 1 displays XRD spectra of the RT-AZO films deposited onto bare glass at a sputtering pressure of 0.13 – 20 Pa. The full width at half maximum (FWHM) of the obtained XRD peaks and the average grain size are summarized in Table I. The average grain size in the deposited RT-AZO film was determined by employing the Debye-Scherrer's equation [33]:

$$D = \frac{0.9\lambda}{\beta \cos\theta}, \quad (1)$$

where D, λ, θ, and β are the grain size, Cu-Kα wavelength, Bragg diffraction angle, and FWHM of the peaks, respectively.

As depicted in Fig. 1 and Table I, the peak intensity of the (002) plane, indicating the c-axis orientation, increased as the growth pressure was adjusted from 0.13 to 20.00 Pa, the FWHM decreased, and the c-axis average crystalline size accordingly increased from 15.35 to 23.19 nm. In contrast, no peaks were observed in the (100) plane at growth pressure of 0.13, 6.67, and 13.3 Pa, respectively. Banerjee *et al.* [34] reported that the growth direction consists of alternate planes or rows of Zn^{2+} and O^{2-} ions due to the introduction of Al^{3+} dopant ions. Al^{3+} ions disturb the charge neutrality of the (100) plane, thereby affecting its surface energy and causing its preferential growth. In the absence of the (100) direction, the charge-neutral surface does not consist of alternating rows of ions on the surface. On the contrary, the RT-AZO film deposited at 20 Pa shows a strong (100) peak at 2θ = 32.94°, which indicates the a-axis grain becomes preferred with a grain size of 46.88 nm, likely because impurities could impede the growth of the c-axis during deposition at high growth pressure [35,36]. The appearance of a-axis peaks at a pressure of 20 Pa indicates that the crystal orientations of the RT-AZO film are influenced by working pressure conditions, which are related to particle energy or the thin film growth rate. The structure zone diagram (SZD) model proposed by Kluth *et al.* [37] suggests that increasing sputtering pressure should lead to greater alignment of ZnO crystallites. However, this model does not explicitly account for the impact of plasma and ion effects on film growth. At low pressures, sputtered particles are more energetic and may introduce defects into the lattice, hindering columnar growth by bombarding the film surface. In this scenario, a fine-grained, nanocrystalline film with a preferred orientation is expected based on the SZD proposed by Refs. [38,39]. Conversely, at higher pressures, the energy distribution of sputtered particles shifts towards lower energies. Additionally, the shadowing mechanism, which promotes rougher surfaces but not larger structures, is less effective at higher pressures due to increased collisions experienced by sputtered atoms and ions before reaching the growing surface. Consequently, the growth of a-axis peak crystallites is affected, as indicated by the 20 Pa data in Fig. 1.

Table I. FWHM, and average grain size of the RT-AZO films under different sputtering pressures.

		0.13 Pa	6.67 Pa	13.3 Pa	20 Pa
RT-AZO (002)	FWHM (deg.)	0.54	0.40	0.38	0.36
	Grain size (nm)	15.35	20.57	21.83	23.19
RT-AZO (100)	FWHM (deg.)	·	·	·	0.18
	Grain size (nm)	·	·	·	46.88

Figure 2 displays the EDS results of as-deposited RT-AZO films under various sputtering pressures. Changes in the sputtering pressure strongly affect the elemental composition of the RT-AZO films. The oxygen atomic fraction increased from 43.5 to 53.4 %, while the Zn atomic fraction decreased from 55.0 to 44.6 % as the operating pressure was increased from 0.13 to 20 Pa. The Al atomic fraction increased from 1.43 to 2.90 %, which is slightly higher than in the

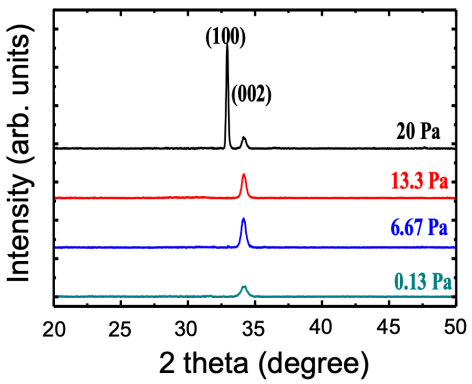


Figure 1. XRD peaks of the RT-AZO films deposited under different sputtering pressures.

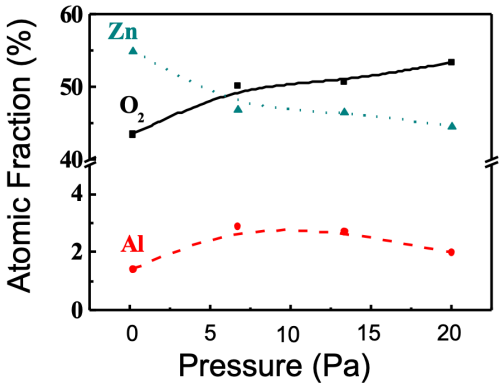


Figure 2. EDS results of the RT-AZO films deposited under different sputtering pressures.

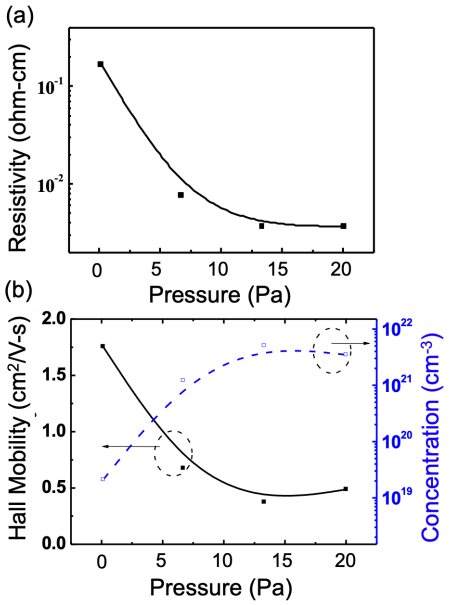


Figure 3. Electrical characteristics of the RT-AZO films deposited under different sputtering pressures. (a) Resistivity and (b) Hall-mobility and carrier concentration.

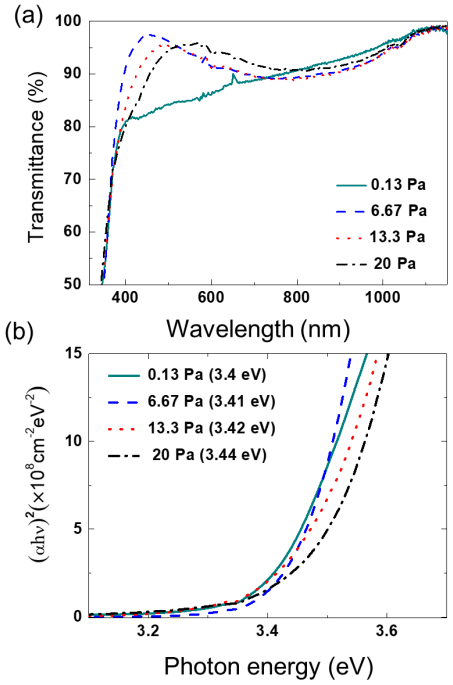


Figure 4. Optical properties of the RT-AZO films deposited under different sputtering pressures. (a) Transmittance and (b) band gap.

pristine target, as the operating pressure was increased from 0.13 to 6.67 Pa, and it then decreased to 2 % when the pressure was raised to 20 Pa. The change of the elemental composition according to the sputtering pressure can be attributed to partial re-evaporation of Zn atoms that arrived at the surface of the target substrate [40]. The backscattering of the sputtered Zn atom is attributed to the decrease in the mean free path [41,42]. In other words, the change in concentration of all elements might be caused by the desorption of both Zn and Al atoms [43]. As a result, the carrier concentration increases with a deposition pressure of 13.3 Pa and then decreases towards a higher deposition pressure of 20 Pa. This, in turn, leads to a decrease in Hall mobility as shown in Fig. 3(b). We will illustrate this further in Fig. 3(b).

The influence of the operating pressure on the electrical properties of the RT-AZO films is depicted in Fig. 3(a). The film resistivity decreased by a significant difference, from 1.7×10^{-1} to 3.7×10^{-3} Ω·cm, as the chamber pressure was increased from 0.13 to 20.00 Pa. A few reports have observed an increase in resistivity as a function of chamber pressure, but those experiments were conducted at lower chamber pressures than applied in our study [44,45]. Numerous studies reported that a decrease in resistivity with increasing sputtering pressure (0.13 to 13.3 Pa) is related to RT-AZO films predominantly oriented along the (002) direction, known as *c*-axis oriented films [46]. However, it is noteworthy that the RT-AZO films deposited at 20 Pa predominantly showed orientation along the (100) direction, known as the *a*-axis orientation [47]. The low resistivity for the (100) direction of the RT-AZO film was a result of growth parallel to the substrate surface for the grains of thin films and the observed dominant diffraction peaks of the (100) direction [48]. In the case of applying sputtering pressure of 20 Pa, the average grain size was increased, and the electron movement path thereby was optimized and then lowered for the resistivity accordingly.

Figure 3(b) shows the influence of the sputtering pressure on electrical properties of the RT-AZO (e.g., Hall mobility and carrier concentration). When the operating pressure was increased from 0.13 to 13.30 Pa, the carrier concentration increased from 2.1×10^{19} to 5.04×10^{21} cm^{-3}. A further increase in operating pressure to 20 Pa slightly decreased the carrier concentration to $3.47 \times 10^{21} cm^{-3}$. However, the change in Hall mobility is not significantly sensitive to variations in pressure. This may be due to the increase in the free electrons generated by Al substitution in the RT-AZO film [48,49].

The transmittance of the RT-AZO films at various sputtering pressures is presented in Fig. 4(a). As the sputtering pressure was increased, the transmittance of the RT-AZO films in the visible range (wavelength of 550 nm) increased from 85 to 95 %. Figure 4(b) displays the optical bandgap (E_g) of the RT-AZO films deposited at various sputtering pressures. To determine the E_g of the RT-AZO films, we utilized Eqs. (2) and (3) [49].

$$T = (1 - R)^2 \exp(-\alpha d) \quad (2)$$

$$\alpha h\nu = D(h\nu - E_g)^n \quad (3)$$

where T, R, and d denote the films' optical transmittance, reflectance, and thickness, respectively. By combining models (i.e., the Davis, Mott, and Tauc models) in the high absorbance region, the E_g of the RT-AZO films is estimated. In Eq. (3), $h\nu$ and D denote the photon energy and the constant of proportionality, respectively. Here, the value of n is set to 0.5 because of the direct transition nature of AZO [35]. In Fig. 4(b), the energy axis displays linear extrapolation of $(\alpha h\nu)^2$ with respect to $h\nu$, which determines the E_g of the RT-AZO films. The E_g of the RT-AZO films increased from 3.4 to 3.44 eV when the sputtering pressure was increased from 0.13 to 20 Pa. This increase in the E_g despite the increase in the sputtering pressure might be caused by the Burstein-Moss band-filling phenomenon [50–53].

4. Conclusions

In conclusion, we have investigated the relationship between sputtering pressure and resistivity saturation in AZO films prepared by RF magnetron sputtering at RT. An XRD analysis revealed that the RT-AZO film deposited at a sputtering pressure of 20 Pa has a predominant orientation along the *a*-axis (100) direction, resulting in a low resistivity of 3.7×10^{-3} Ω·cm and a high transmittance of 95 % in

the visible range. We believe that our results can be used to produce highly conducting and transparent AZO films, which are promising candidates for applications in low-cost transparent electronics.

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Conflicts of Interest

The authors declare no conflicts of interest.

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